Fairness Issues in Data Mining

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***Abstract*—Word embeddings encode semantic structure in vector space and are foundational to modern natural language processing systems. While static embeddings assign one vector per word type, contextual embeddings vary by sentence usage, allowing models such as BERT to capture nuanced meaning. This project investigates gender bias in contextual embeddings across distinct literary contexts: Indian literature and Western literature (split by author gender). Using `bert-base-uncased`, we extract contextual representations, compute word centroids from the last four hidden layers, and quantify bias with a Mean Average Cosine (MAC) score relative to male and female target sets. Results show that bias patterns differ significantly across corpora, indicating that representational bias is context-sensitive rather than universal. These findings motivate culturally grounded fairness evaluation before deploying language technologies.**

I. Introduction

Word embeddings map words into continuous vector spaces where geometric proximity reflects semantic relatedness. Early static embeddings cannot distinguish context-dependent meanings, whereas contextual models produce token representations conditioned on surrounding text. BERT generates contextual embeddings via deep bidirectional Transformer layers, where each layer refines semantic and syntactic information.

Fairness in embeddings matters because these representations are reused in downstream tasks and can propagate harmful associations. Biased embeddings can produce representational harms (e.g., stereotyping in translation or dialogue) and allocational harms (e.g., unequal ranking in hiring pipelines). Therefore, investigating embedding-level bias is a crucial step for responsible ML.

The objective of this study is to evaluate gendered representational bias not only in aggregate, but also across cultural and demographic contexts. Specifically, we compare Indian literary text with Western literary text and further split the Western text by female versus male authorship.

II. Experimental Setup: Data and Model

A. Model

All experiments use Hugging Face's `bert-base-uncased`, which outputs 768-dimensional hidden states for each token position.

B. Data

Raw text is collected from Project Gutenberg and organized into two primary corpora:

- **Corpus 1 (Indian Literature):** Texts representing the Indian cultural context.
- **Corpus 2 (Western Literature):** 19th and 20th-century Western texts, split into:
  - **Female-author sub-corpus**
  - **Male-author sub-corpus**

III. Methodology

A. Contextual Embedding Extraction

For each target or attribute word, we collect contextual sentences from each corpus. Each sentence is tokenized and passed through BERT. For each occurrence of a word, we average the hidden states from the **last four layers** to capture higher-level semantic information. We then mean-pool across all contexts of that word to obtain one centroid vector per word.

B. Target Sets (Gender Anchors)

We define two anchor sets:

- **Male:** man, men, male, he, him, his, boy, father, son, brother, husband, king
- **Female:** woman, women, female, she, her, hers, girl, mother, daughter, sister, wife, queen

C. Attribute Sets

We evaluate four semantic domains:

- Science
- Arts
- Professions
- Stereotype Traits

D. Bias Metric: Mean Average Cosine (MAC)

Let a be an attribute-word centroid, M the male target set, and F the female target set. We compute:

$$\text{Bias}(a) = \frac{1}{|M|} \sum_{m \in M} \cos(a, m) \; - \; \frac{1}{|F|} \sum_{f \in F} \cos(a, f). \quad (1)$$

A positive score indicates a stronger male association; a negative score indicates a stronger female association.

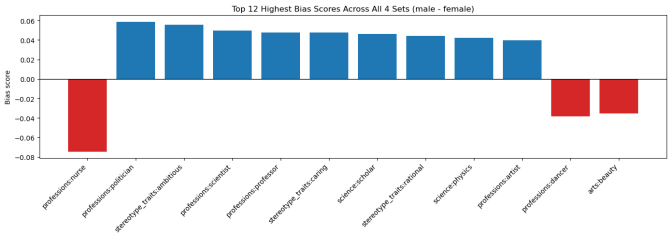

Fig. 1. Top 12 words exhibiting the highest absolute Mean Average Cosine (MAC) bias scores (**Indian Literature Context**). Positive values indicate a stronger mathematical association with male target concepts, while negative values denote a stronger association with female target concepts.

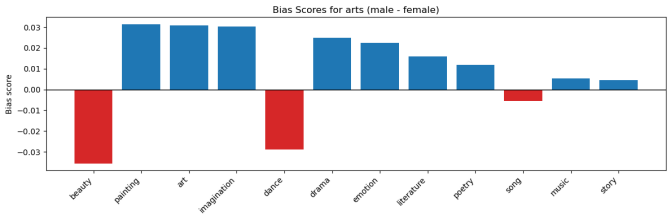

Fig. 2. Bias distribution within the "Arts" domain for the Indian Literature corpus. Unlike Western texts where arts are broadly female-coded, this corpus exhibits a strong male bias for intellectual and formal arts (e.g., *painting*, *imagination*, *literature*), restricting female associations primarily to aesthetic and performative concepts (e.g., *beauty*, *dance*).

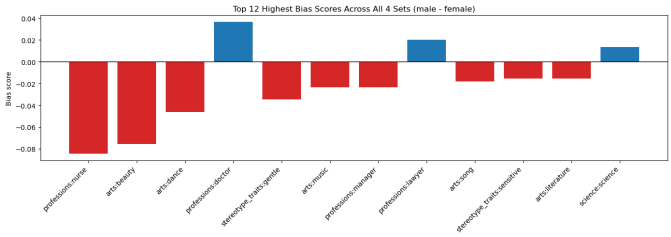

Fig. 3. Top 12 words exhibiting the highest absolute bias scores (**Western Literature Context**). The data reveals extreme semantic polarization across domains: authoritative occupational roles and STEM fields (*doctor*, *lawyer*, *science*) skew heavily male (positive), while supportive roles, aesthetic arts, and specific traits (*nurse*, *beauty*, *gentle*) skew female (negative).

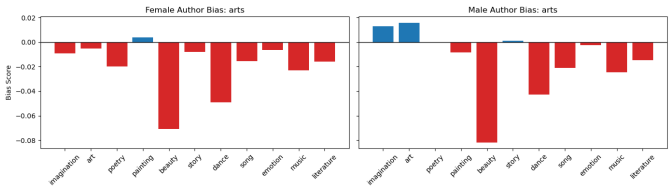

Fig. 4. Comparative bias within the "Arts" domain between Female authors (left) and Male authors (right) in the Western corpus. While both demographics broadly associate the arts with the female semantic space, Male authors distinctly reserve high-level creative concepts (e.g., *imagination*, *art*) for the male domain, illustrating an intra-domain gender divergence based on the author's gender.

IV. Results and Findings

A. Indian Literature

Indian-literature results show clear category-level asymmetry. Profession words are predominantly male-leaning, with the strongest male associations including *politician* ($+0.0587$), *scientist* ($+0.0498$), and *professor* ($+0.0476$). The largest-magnitude bias overall is *nurse*, which is strongly female-leaning at -0.0745. In stereotype traits, *ambitious* shows a strong male-leaning signal ($+0.0555$). In arts, intellectual and literary terms such as *painting*, *imagination*, and *literature* tend to lean male, while performative aesthetic terms such as *dance* and *beauty* are comparatively female-leaning. Together, these findings indicate that the Indian corpus encodes both occupational and trait-based gender structuring, with one of the strongest effects centered on profession terms.

B. Western Literature: Female vs. Male Authors

The Western split-corpus analysis reveals two patterns. First, an **agreement but magnitude gap**: both author groups encode similar stereotypes, but with different strengths. For example, *nurse* is female-leaning in both groups, but much more strongly in male-author text (-0.1014) than in female-author text (-0.0705). The largest absolute bias in the female-author corpus is *beauty* (-0.0708), while the largest absolute bias in the male-author corpus is *nurse* (-0.1014).

Second, we observe a **disagreement in bias**. The word *strong* flips direction across sub-corpora: female-author text is slightly female-leaning (-0.0061), while male-author text is male-leaning ($+0.0201$). Similar directional disagreements appear for *imagination* (-0.0092 vs. $+0.0128$). The largest cross-group divergence ($|\Delta|$) is for *manager* ($\Delta = +0.0331$), followed by *nurse* ($\Delta = -0.0309$) and *strong* ($\Delta = +0.0261$). These results suggest that author-demographic context changes not only bias magnitude, but in some cases even its direction.

V. Discussion and Cross-Cultural Analysis

The central finding is that embedding bias is not universal; it is shaped by local cultural and historical context. In our Western corpus, arts-related terms are largely female-leaning, while in the Indian corpus arts terms are more male-leaning for intellectual/literary words, with aesthetic terms (e.g., beauty, dance) leaning female. This divergence suggests that fairness assumptions cannot be transferred unchanged across regions.

From a deployment perspective, global NLP systems require localized fairness diagnostics. A model that appears balanced under one corpus may encode materially different stereotypes under another, potentially producing inequitable outcomes across user populations.

VI. Conclusion

We successfully extracted contextual representations from `bert-base-uncased` and quantified systematic gender associations in a 768-dimensional embedding space using MAC bias scores. The analysis demonstrates robust cross-corpus and cross-demographic variation in bias expression, highlighting the need for context-aware fairness evaluation.

For future work, we plan to evaluate mitigation methods including Counterfactual Data Augmentation, Algorithmic Regularization during fine-tuning, and Geometric Hard Debiasing in representation space.

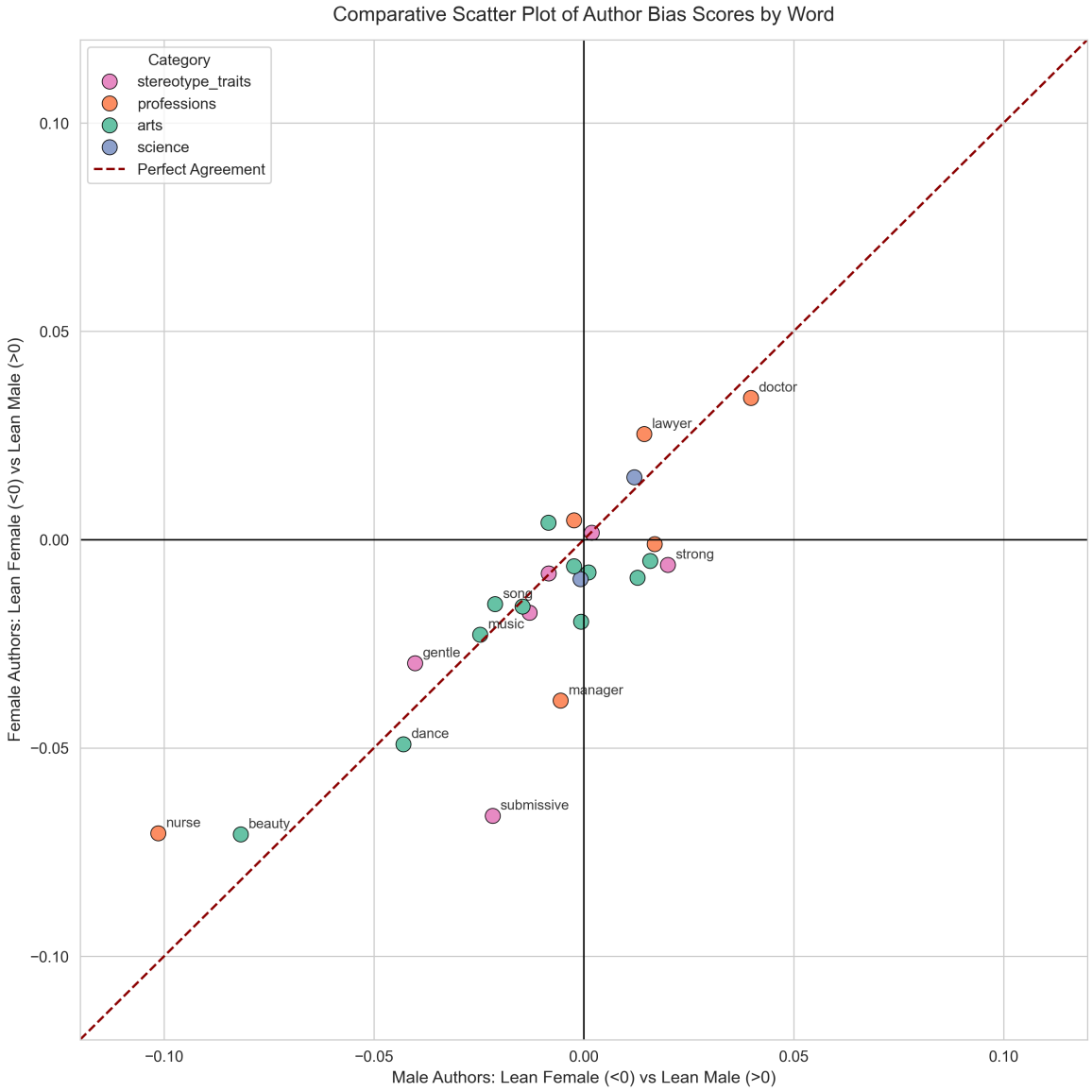

Fig. 5. Comparative scatter plot of word embedding bias scores, illustrating the difference in gender associations learned from Male-authored (x-axis) versus Female-authored (y-axis) texts. The dashed diagonal line denotes perfect agreement between the two corpora. Deviations from this line highlight divergence in cultural encoding; for instance, Female authors push *manager* and *submissive* further into the female semantic space compared to Male authors.

Code Availability

The complete pipeline implementation, data processing scripts, and visualization notebooks for this project are available in the following GitHub repository:
https://github.com/maverick-0215/Fairness-issues-in-BERT

References

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